

DATE: 09/29/2026
TO: Kid Kidd
FROM: Paa Woode
SUBJECT: Project 2

1 INTRODUCTION

As engineers, the documentation of a project is just as crucial as implementing the project itself. The purpose of this project is to modify the design documented in project 1 so that it uses a polled, round-robin cyclic executive task manager that dispatches tasks through a function pointer array. The Customer Requirements, High-level Technical Requirements and Education Constraints are unchanged. This document is a new revision of project 1. It includes an Engineering Change Note (ECN1.1) in Section 8 and marks all changes to project 1 with the date of the change.

2 PROBLEM STATEMENT

There are 3 parts to this assignment, mirroring project 1. In Part I, the Customer Requirements and derived High-level Engineering Requirements are unchanged and are incorporated by reference. In Part II, the testing requirements of project 1. are retained and extended with tests needed to evaluate the cyclic executive. In Part III, the system is redesigned around a cyclic executive, built and tested.

3 EQUIPMENT

- Arduino Uno R3 development board or clone, e.g. the ELEGOO Uno R3
- LEDs, qty 2
- 200 Ω ¼ Watt resistors, qty 2
- 10k Ω ¼ Watt resistor, qty 1
- Arduino IDE 2.3.3 or better
- Laptop or another computer to run the Arduino IDE
- Cable, USB 2.0 female type B to female USB 2.0 type A

4 PART I CUSTOMER REQUIREMENTS -> HIGH-LEVEL ENGINEERING REQUIREMENTS

PROBLEM TO BE SOLVED: From the Customer's requirements in Table 1, derive the high-level technical requirements for the system being designed. These requirements are written in the Customer's language and understandable by the Customer. Typically, these requirements are not specific enough to be used for engineering purposes.

Table 1. Customer Requirements

ECO 1.0	Implement a cyclic executive task manager based upon a function pointer table to dispatch the asynchronous tasks created in project 1.
ECO 1.1	The testing and validation requirements must contain those defined in project 1.
ECO 1.1.1	The testing and validation requirements must be extended with any additional tests necessary to evaluate the modified system.
ECO 1.2	The final report must be a new revision of the formal design document produced in project 1, noting all changes.

5 PART II: DERIVE TESTING REQUIREMENTS

PROBLEM TO BE SOLVED: Given the high-level technical requirements (HL. #) and educational constraints (EC.#) below – and different from the one’s you derived in Part II – derive the testing requirements (labeled T.#) that will allow the company to prove to the customer that The Company has satisfied the contract.

Table 2. High Level technical requirements

Table 3. Education Constraints

EC.1	The Developer must use the version of gcc used by Arduino IDE 2.3.3 or later for the AVR ATmega328p to cross compile any program used by the System
EC.2	The Developer must have any software used in The System hosted on a GitHub repository. This software does not need to include libraries, macros or other code not developed by The Developer.
EC.2.1	1 The Evaluator must be able to (a) download from the repository any software, documentation and other material needed for their evaluation, (b) cross compile any software, and (c) run that software on The Reference Platform.
EC.3	LEDs must be connected to Arduino digital Pins 2 & 3.
EC.4	The Evaluator must be able to successfully have the system perform the dialog in Table 4
EC.5	The Developer must do bare metal programming, i.e. they cannot use any 3rd party multitasking libraries or programs such as FreeRTOS.

Table 4. Example Dialog (*Red* are the user inputs).

Serial Port I/O	Notes
What LED (1 or 2) 2	
What interval (in msec)? 600	LED2 starts blinking at an interval of 300ms on and 300ms off. LED1 is unaffected.
What LED (1 or 2) 1	
What interval (in msec)? 1600	LED1 starts blinking at an interval of 800ms on and 800ms off. LED2 is unaffected.
What LED? (1 or 2)	Waiting for next LED# interval pair

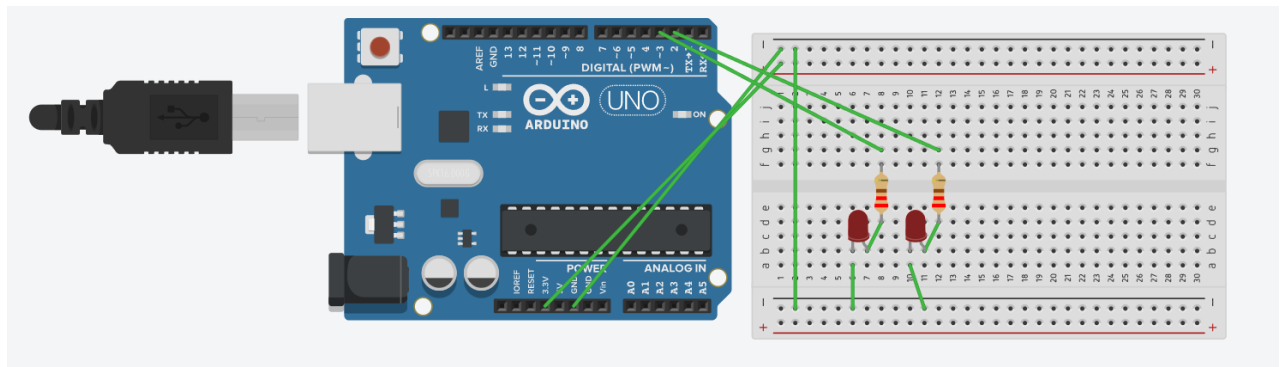
T.1	The Evaluator must visually confirm that two LEDs are present and wired, and that each LED illuminates independently when its corresponding digital pin is driven HIGH.
T.2	The Evaluator must confirm, by inspection of the physical wiring or schematic, that LED1 is connected to digital Pin 2 and LED2 is connected to digital Pin 3.
T.3	The Evaluator must open the Arduino IDE serial-monitor and confirm that prompts and responses matching Table 4's dialog are visible.
T.4	With both LEDs blinking at two different, previously set intervals, the Evaluator must send a new interval to one LED via the serial-monitor, and confirm that the unaffected LED's blink rate does not change or pause during or after the input.
T.4.1	The serial monitor must remain responsive to further input immediately after the change.
T.5	The Evaluator must enter a specific interval (in msec) via the serial monitor for a given LED and measure that LED's actual on/off period using a video timestamp.
T.6	With one LED set to a fixed interval, the Evaluator must send a new interval to the <i>other</i> LED and confirm the first LED's period does not change and continues blinking at its original rate.
T.7	The Evaluator must confirm, via the build output or project configuration file, that the tool chain used to compile the software is the AVR gcc distributed with Arduino IDE 2.3.3 or later.
T.8	The Evaluator must clone the Developer's GitHub repository onto a machine with only Arduino IDE 2.3.3 or later
T.9	The Evaluator must execute the exact dialog sequence shown in Table 4 via the serial monitor and confirm that:

T.9.1	Each prompt text matches the example dialog.
T.9.2	LED2 begins blinking at a 300 ms on 300 ms off rate after the 600ms inputs
T.9.3	LED1 begins blinking at an 800 ms on 800 ms off rate after the 1600ms input, without affecting LED2.
T.9.4	After both inputs, the serial-monitor returns to prompting What LED? (1 or 2
T.10	The Evaluator must inspect the source code and confirm no 3rd-party multitasking library.
T.11	The Evaluator must inspect the source code and confirm that the tasks are dispatched from an array of function pointers, called in a fixed round-robin order from loop ().
T.12	The Evaluator must inspect the source code and confirm that no task contains a blocking call (delay (), delay Microseconds (), Serial.parseInt(), Serial.readString(), or a busy-wait loop).
T.13	The Evaluator must re-execute T.4, T.5, T.6 and T.9 of project 1 on the cyclic executive build and confirm the results are identical to those of project 1 (regression test).
T.14	With both LEDs blinking, the Evaluator must enter a partial input (e.g. "1" with no interval) and confirm that both LEDs continue blinking undisturbed while the System waits for the remainder of the input.
T.15	The Evaluator must leave the serial-monitor idle for 60 seconds with both LEDs blinking, then enter a new LED/interval pair, and confirm the System accepts it and neither LED paused.
T.16	The Evaluator must measure the on/off period of each LED over at least 10 cycles using a video timestamp and confirm it is within $\pm 5\%$ of the entered value.

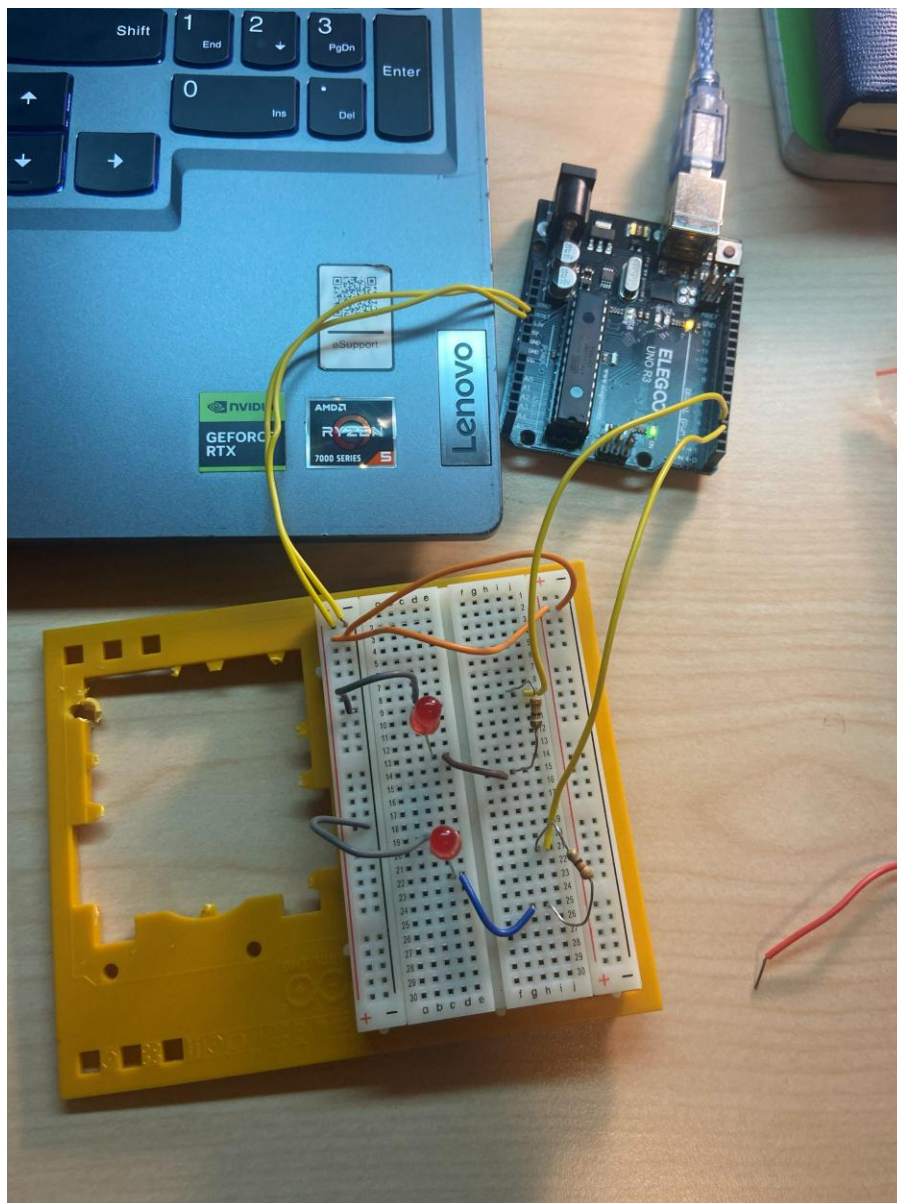
6 PART III: DESIGN, BUILD AND TEST SYSTEM

Schematic:

This schematic was done with Tinker CAD circuits. There were a couple of assumptions that LEDs are standard red LEDs. The board is an Arduino UNO and LED 1 is connected to pin 2 and LED is connected to pin 3.



Hardware Design:



Resistor Values: 330 ohms. I had to use a 330-ohm resistor because I had no 200-ohm resistor.

7 ENGINEERING CHANGE NOTE

PROJECT	PROJECT-ASYNC (Project #1)
ORIGINATION	Prof. Kidd, ECO 0.1
PURPOSE OF CHANGE	Cyclic executive dispatch
DOCUMENT NUMBER	ECN1.1
DATE OF PUBLICATION	09/29/2026
LEAD ENGINEER	Kid Kidd, UMBC
TEAM LEAD	Paa Woode
EXECUTIVE SUMMARY	Per ECO 0.1, the design of [1] was modified to use a polled, round-robin cyclic executive that dispatches tasks through a function pointer array. The serial dialog was rewritten as a non-blocking state machine, and LED blinking was split into two independent tasks. Customer requirements C1 through C6 and high-level requirements HL.1 through HL.6 are unchanged. Testing requirements T.1 through T.10 is retained and T.11 through T.16 are added. Hardware is unchanged.
HIGH-LEVEL DESCRIPTION	The task management of the design in [1] is replaced with a function pointer based cyclic executive.
REASON FOR CHANGE	Results in a more reliable design and reduces labor and design costs of future extensions (ECO 0.1).
CHANGED DOCUMENTS	09/29/2026, Project 1 Section 6, Software: the previous task handling (loop() calling the blink and Serial.parseInt() routines directly) replaced by "Tasks are dispatched in fixed round-robin order from an array of function pointers in loop()." 09/29/2026, Project 1 Section 5: T.11 through T.16 added. 09/29/2026, Project 1 Section 7: "Testing (T.1–T.10) confirmed the System correctly reproduces the example dialog" revised to reference T.1 through T.16 (see Section 7 of this document).

ADDED DOCUMENTS	This revision (ECN1.1 included); cyclic executive source file; tests T.11 through T.16.
REMOVED DOCUMENTS	None.
REPLACED DOCUMENTS	The software design of project 1 is replaced by Section 6 of this revision.
ANY ADDITIONAL DESIGN CONSIDERATIONS	AVR gcc from Arduino IDE 2.3.3 or later (EC.1). No 3rd-party multitasking libraries (EC.5). LEDs remain on Pins 2 and 3 (EC.3). Serial monitor line ending must be Newline or Both NL & CR. Resistors remain 330 Ω per [1].

8 CONCLUSIONS

This project derived high-level requirements from the customer's requirements (Part I), derived testing requirements from a separate set of high-level requirements and constraints (Part II), and used those to design, build, and test a working System (Part III). The completed System independently blinks two LEDs on digital Pins 2 and 3, with intervals set via the serial-monitor without one LED's timing affecting the other. Testing (T.1–T.10) confirmed the System correctly reproduces the example dialog and satisfies HL.1–HL.6, with the only deviation being the use of 330 Ω resistors in place of 200 Ω due to parts availability, which dimmed the LEDs slightly but did not affect functionality. Overall, the System satisfies all specified requirements and constraints.